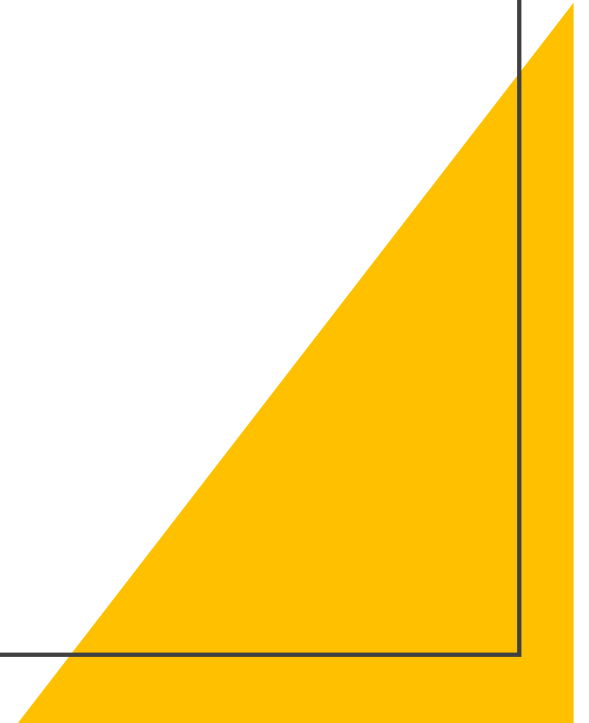


Data Analytics Technology Decision Support Systems

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Agenda

- Decision support systems
- Big Data

Decision Support (1)

- It is important that MANAGERS and information technology professionals understand data-driven **decision support systems** (DSS) and how these systems can provide business **intelligence** and performance monitoring.
- Data-driven DSS are one of the main types of computerized decision support systems and the capabilities of these systems will vary depending on specific implementations.
- **It is very important to know what questions the business intends to answer!**

Decision Support (2)

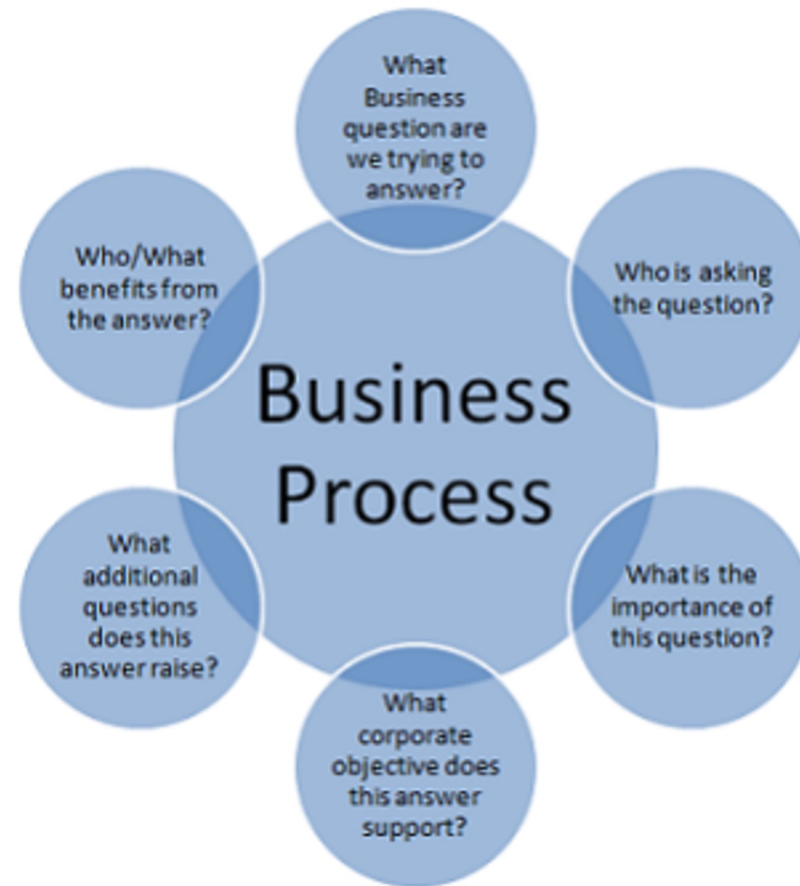
- **Better Decisions:** obtaining and extracting information from new data sources to improve decision quality
- **Faster decisions:** enabling real-time data capture and analysis
- **Decisions that make a difference:** focus data collection and analysis efforts on areas that allow for true differentiation

Decision Support (3)

**Data analytics + business intelligence
= reduced waste**

“The most dangerous kind of waste is the waste we do not recognise” —
Shigeo Shingo

Decision Support Systems



Why are DSS important?

90% of organisations fail to execute their strategies successfully

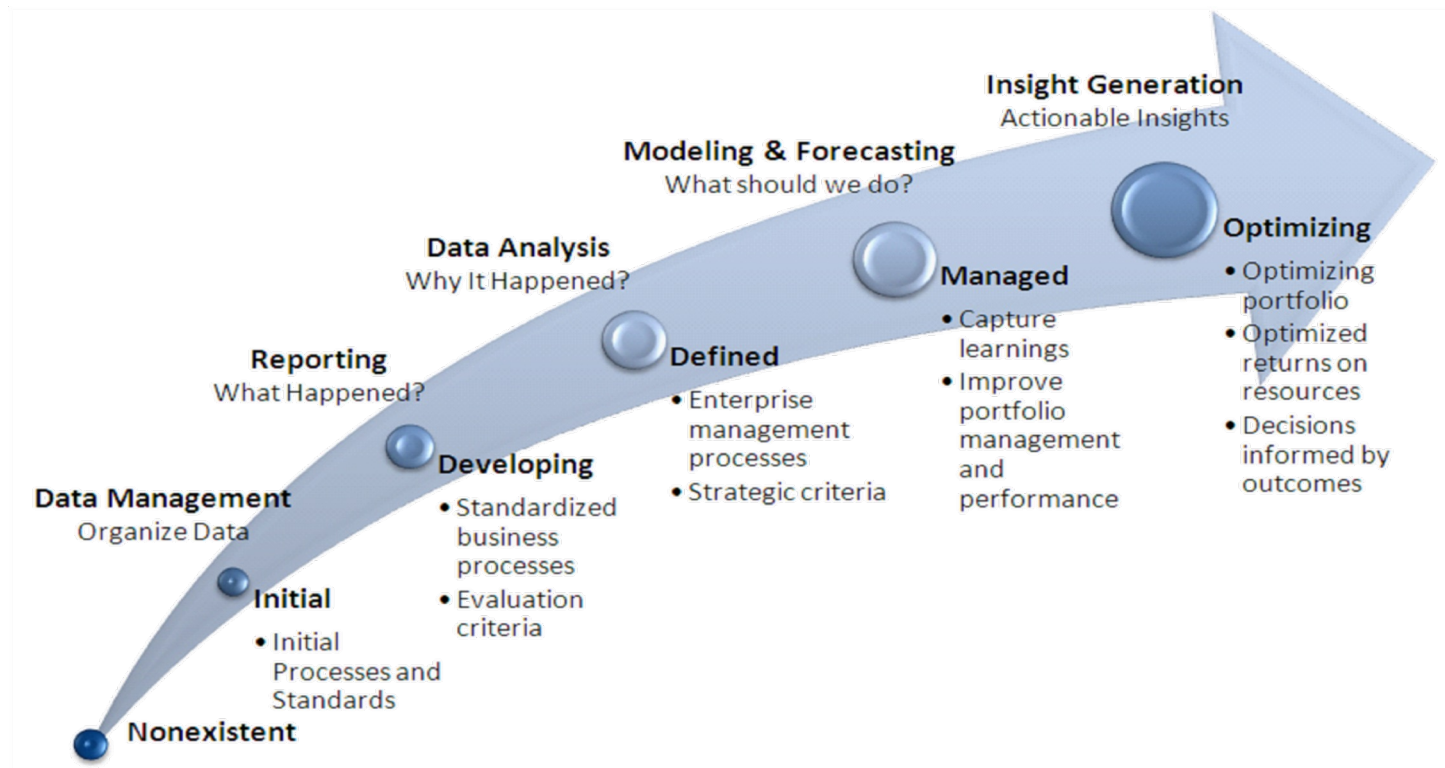
95% of a typical workforce doesn't understand how the organisational goals relate to their job.

86% of teams spend less than one hour a month discussing strategy

60% of organisations don't link their strategies to plans and budgets

Source: Palladium Group/BSCOL

Maturity and DSS



How are good management decisions made?

- Flipping a coin
- Insight or “that feeling”
- Experience and/or expert advice
- Imitation
- Industrial espionage and secrets
- By the “book”
- ... and many others

There is no silver bullet!
Everything here might
get the job well done!

One more question...

What is the most important artifact for doing good management?

Having a GOAL will help you devise
(revise) a STRATEGY

Now that you have a goal... Decision support

- How do you get there?

- What is the right (best) strategy?
 - Flip a coin, insight, experience, etc
- How do you know if your strategy is working?
- How do you know if your strategy is being applied correctly?
 - DATA
 - DATA
 - INFORMATION

Key Performance Indicators (KPIs)

The diagram illustrates a feedback loop. A large grey arrow originates from the 'Key Performance Indicators (KPIs)' box, points left, and then turns downward to point at the 'How do you know if your strategy is working?' question. A smaller grey arrow points from the 'DATA' and 'INFORMATION' items to the 'Key Performance Indicators (KPIs)' box.

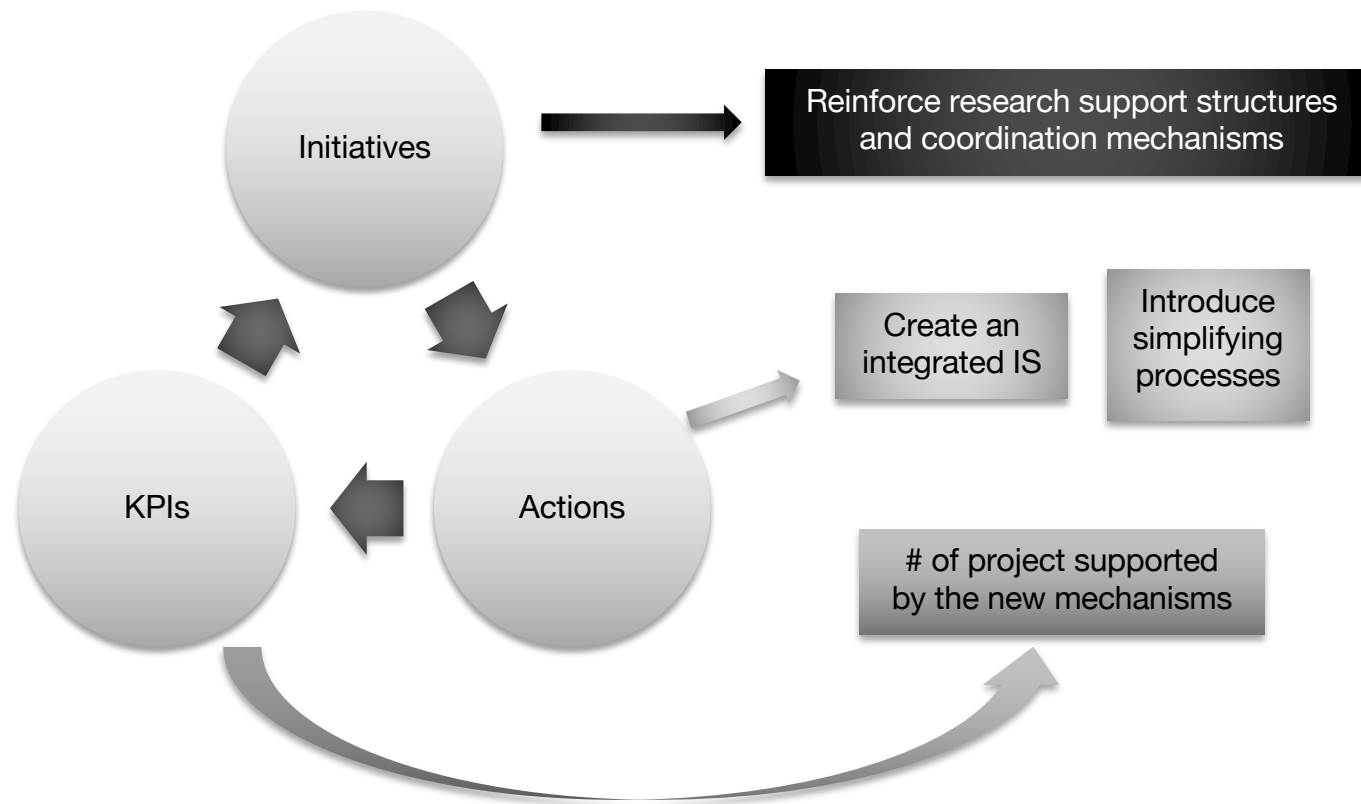
@University of Coimbra

- Strategic Plan
 - 2011-2015
 - 2016-2019
- Areas
 - Research, Teaching, Knowledge Transfer, People, Economic, Infrastructure, Organizational
- KPIs for each area
 - e.g. number of publications by faculty in web of science
- Strategic initiatives and actions
 - e.g., increase participation in research networks
 - e.g., create agents and mechanisms for studying market opportunities for focusing the research offer at the UC

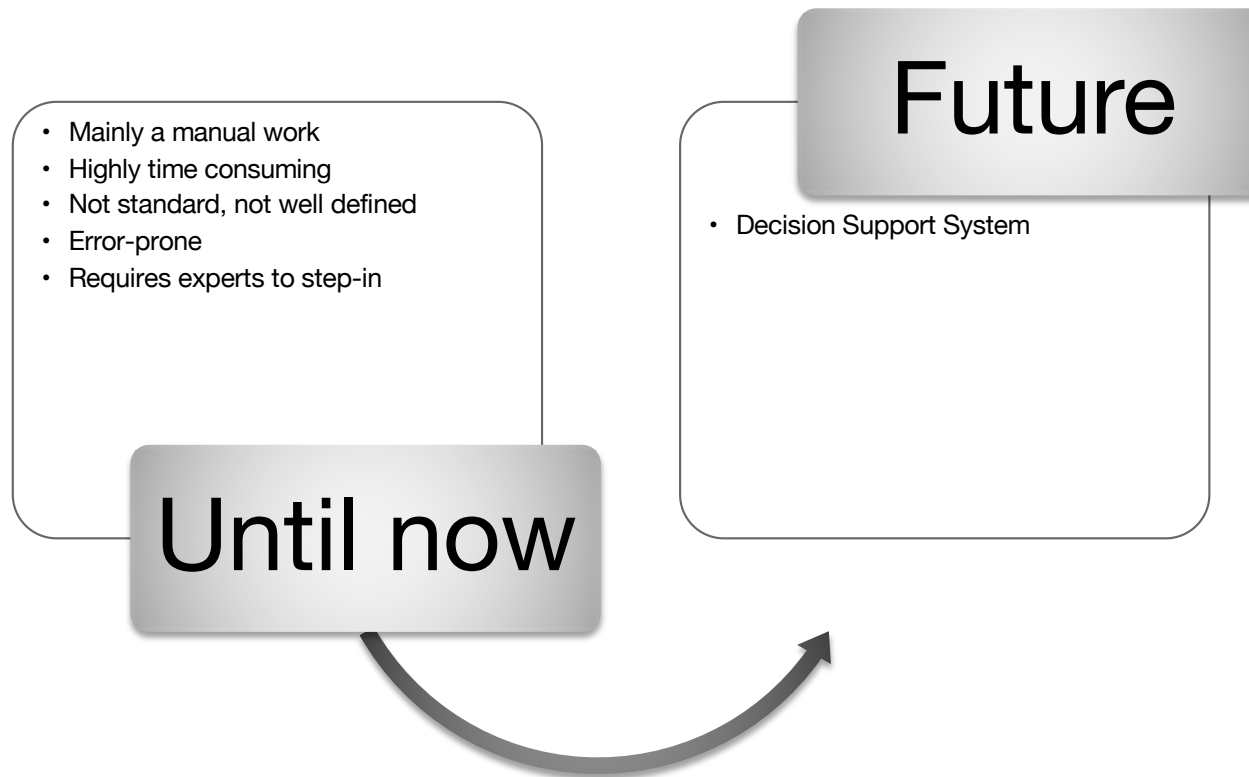
KPIs

- Research
 - # of European projects
- Teaching
 - Growth rate in the # of students
- Knowledge transfer
 - # of projects in consortium
- People
 - # of non-faculty doing training course each year
- Economic
 - Average payment time to suppliers
- Infrastructure
 - Maintenance cost by m3
- Organizational
 - Growth rate in UC brand sales

Initiatives and actions



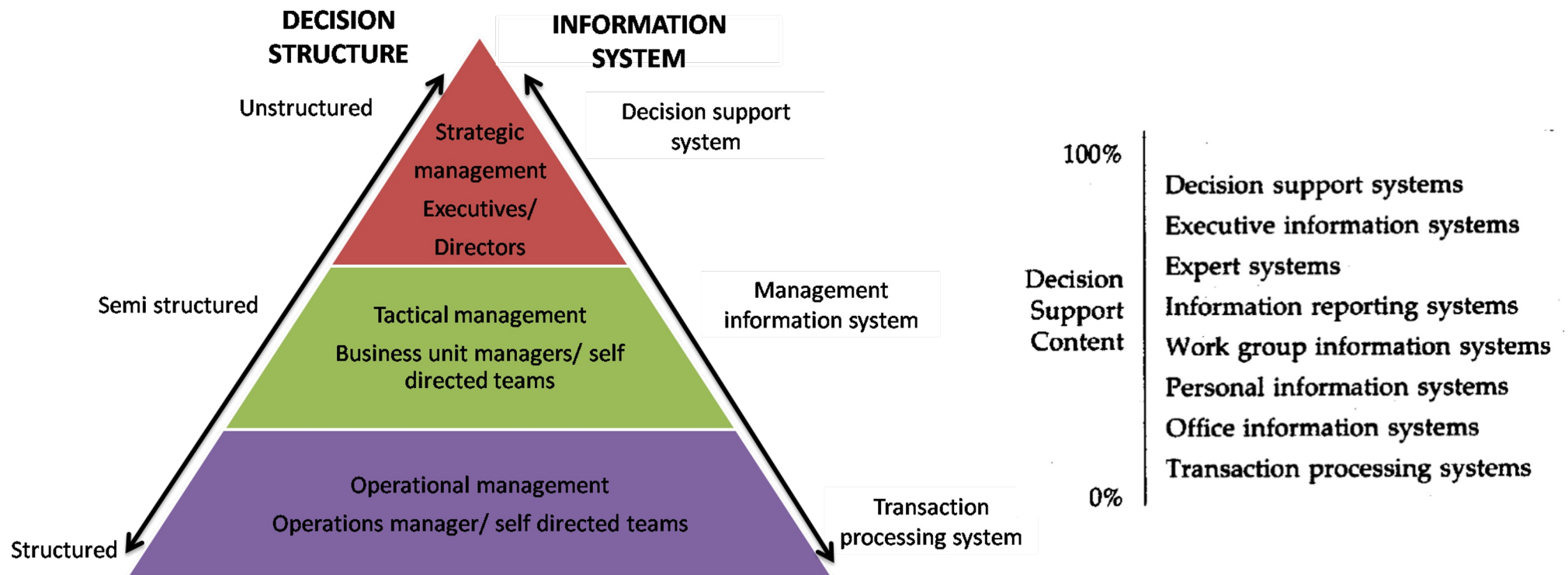
Monitoring



Types of DSS Systems

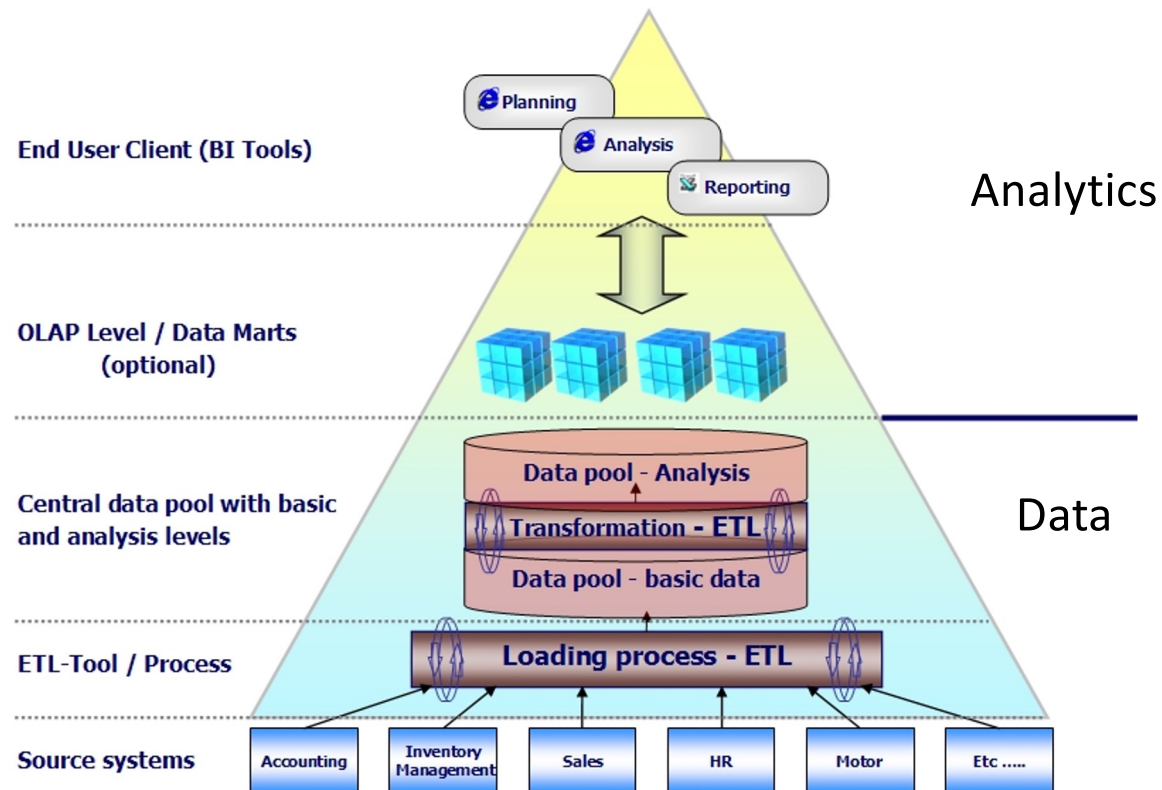
- Model-driven
- Data-driven
- Communication-driven
- Document-driven
- Knowledge-driven

The Pyramid of decision



Handling data

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BIG DATA



- Everyone talks about Big Data
- Nobody knows how to do
- Everyone thinks everyone is using it
- Everyone says they are using it

Big Data: 10 V's

- Very large sets of data that can be analyzed computationally to detect patterns, trends and associations between variables.

Volume

Amount of data generated and stored

Variety

Type and nature of data

Velocity

How quickly data is generated

Variability

Inconsistency in the dataset

Veracity

Variance in the quality of the data generated

Value

Profit of data to the business

Visualization

Ability to show information.

Validity

Adequacy of data to objectives.

Vulnerability

Data security.

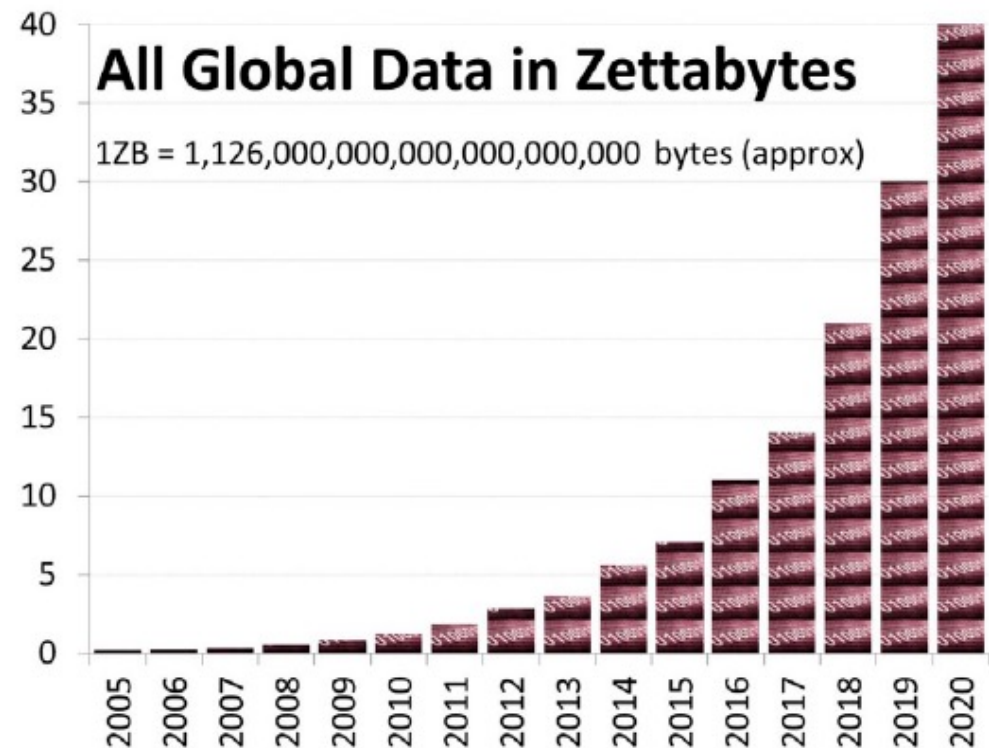
Volatility

When data is no longer relevant

Big Data: Volume

- The amount of data generated every second by machines, networks, humans, sensors, etc. is huge and growing
- Examples:
 - Airbus 10 Terabytes every 20 min, 640 Terabytes in a flight
 - Autonomous cars will generate 2 Petabytes per year
 - Facebook has over 500 Terabytes every day

GB - TB - PB – EB – ZB



The growth in data as seen by United Nations Economic Commission for Europe.

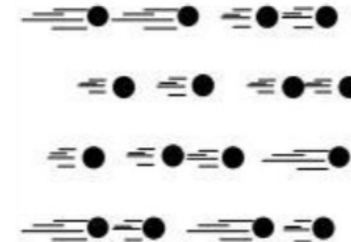
Big data: Variety

- Before, data was structured in DBs, such as numbers, dates, text.
- Today we have data in different formats:
 - georeferenced data
 - sensor data
 - 3D data
 - audio
 - Video
 - Unstructured text, log files
 - Social media



Big Data: Velocity

- The speed of data generation and the speed with which they have to be analyzed is increasing
- Arrival of data in streams leads to fast memory usage instead of disk, or specific disk systems
- IoT often requires real-time reaction, often with machine-to-machine communication processes
- Examples:
 - Stock market data
 - Sensor data logs
 - User data (clicks, games, etc.)

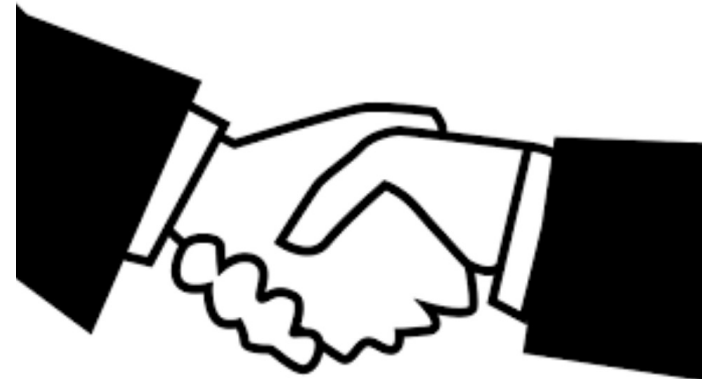


Big Data: Variability

- Refers to changes in concepts (concept drift)
- It is often confused with variety:
 - For example, if a bakery has 10 types of bread, it has a lot of **variety**.
 - On the other hand, if the same type of bread varies in shape/flavor/smell on different days, it will present **variability**

Big Data: Veracity

- It relates to the provenance of the data, it has to do with trust in the sources
- Data veracity analysis helps validate and understand the risks associated with analysis and business decisions based on certain data.
- The data can be:
 - accurate / imprecise
 - Credible / not plausible
 - Objective/subjective
 - Trustworthy / Deceitful



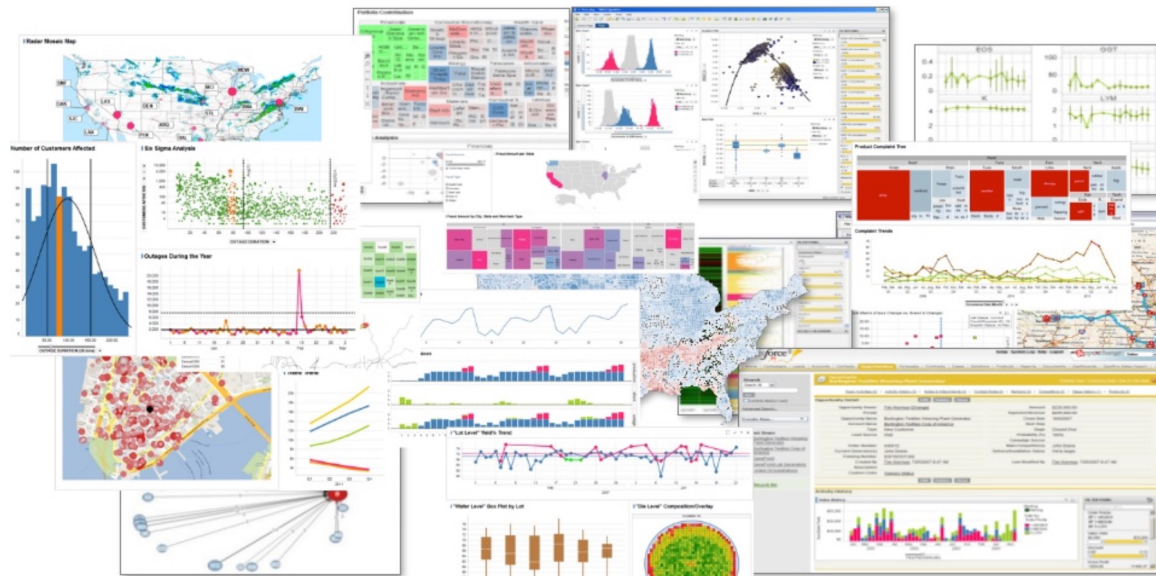
Big Data: Value

- Having a lot of data is only interesting if you can turn it into value
- Examples:
 - Discover a customer's preference or sentiment
 - Present location-based offers
 - Identify a part or equipment that will fail



Big Data: Visualization

- One of the most difficult facets of applying Big Data techniques
- Make the large amount of data easy to understand and read, in order to support/make decisions



Big Data: Validity

- Validity has to do with the suitability of data for the intended use, with data quality and interpretability, i.e., whether the results are relevant to a given problem.
- It depends directly on veracity, because only when veracity is guaranteed can the validity be evaluated.
- Especially relevant in automatic decision making, when it is necessary to be sure that both the data and the analysis are correct.



Big Data: Vulnerability

- Vulnerability is related to security concerns, with unauthorized access
- Recent cases of vulnerability exposures and security breaches are obvious examples of what has to be a concern



Big Data: Volatility

- Volatility is related to how long the data remains valid/necessary
- How long do we have to maintain storage structures for “old” data
- The speed of data generation and its volume are directly linked to volatility, due to the storage and processing effort.



Big Data: Challenges

- Learn to make the right questions

